

Web Development

BRANCH: INFORMATION TECH (IT)

COURSE CODE: IT201

Subject Overview & Scope

This comprehensive document serves as the official compiled study notes for Web Development (IT201) under the Information Tech (IT) department. It is structured into four core university syllabus units covering primary concepts, definitions, formulas, and advanced technical applications for university examinations and placements.

University Course Syllabus & Core Units

Unit 1: Frontend Fundamentals

Introduction to HTML5, semantic elements. CSS3: Box model, selectors, Flexbox, Grid, Responsive design. Javascript ES6+ features, DOM manipulation.

Unit 2: Advanced Frontend Frameworks

React.js: Components, JSX, Props, State, Hook lifecycle (useState, useEffect), Routing (React Router), State Management (Context API, Redux).

Unit 3: Backend Development

Node.js environment, NPM. Express.js: routing, middleware, RESTful API design. JWT authentication, password hashing.

Unit 4: Database Integration & Deployment

MongoDB: schema design (Mongoose), CRUD operations. SQL databases (MySQL). Deployment: Git, cloud hosting platforms.

Flexbox properties (justify-content, align-items). REST: GET, POST, PUT, DELETE mappings.

CRITICAL FORMULA / PRINCIPLE:

Key Formulas & Exam Pointers



